NATIONAL INSTITUTE OF TECHNOLOGY PATNA

Department of Mechanical Engineering

RESEARCH PROJECT REPORT

DESIGN, ANALYSIS AND FABRICATION OF AN EARTH TUBE HEAT EXCHANGER

Submitted in partial fulfilment of the requirements for

Bachelor of Technology in Mechanical Engineering

Submitted by

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Academic Session: 2023–2027

Patna, Bihar, India

EXPANDED REPORT WITH ILLUSTRATIVE EXPERIMENTAL DATA

Numerical observations are synthetic examples, not verified project measurements.

CERTIFICATE

This is to certify that the report entitled “Design, Analysis and Fabrication of an Earth Tube Heat Exchanger” is submitted by Gaurav Kumar (Roll No. 2301089), student of B.Tech Mechanical Engineering, National Institute of Technology Patna, in partial fulfilment of the academic requirements of the programme.

The report presents the design methodology, engineering calculations, simulation workflow and an illustrative experimental-data analysis. Project-specific values must be verified and approved by the project supervisor before final submission.

Project Supervisor: _____

Head of Department: _____

Date: _____ Place: Patna

DECLARATION

I declare that the final version submitted for academic assessment will accurately represent the work actually performed, the simulation models created, the fabrication steps completed and the measurements obtained. Any assumed or synthetic data used to demonstrate calculations will be identified and replaced by verified project records wherever available.

Signature: _____ Name: Gaurav Kumar Roll No.: 2301089

ACKNOWLEDGEMENT

I express my sincere gratitude to my project supervisor and the Department of Mechanical Engineering, National Institute of Technology Patna, for academic guidance and support. I also acknowledge the assistance of faculty members, laboratory staff and colleagues who contributed to discussions on thermal design, CAD modelling, fabrication and testing. This section should be personalised to include only people and facilities that actually supported the project.

ABSTRACT

An Earth Tube Heat Exchanger (ETHE), also known as an earth–air heat exchanger, uses the relatively stable temperature of the ground to pre-condition ventilation air. Air passes through a buried pipe and exchanges heat with the surrounding soil. During warm weather, when the soil is cooler than the incoming air, the system can reduce air temperature; in colder conditions, heat transfer may reverse.

This report develops an end-to-end design and analysis workflow using an illustrative 20 m pipe with an internal diameter of 0.150 m, volumetric airflow of 0.050 m^3/s, inlet air at 38°C, outlet air at 30°C and soil at 25°C. Calculations from these assumed values give a mean velocity of 2.829 m/s, an internal heat-transfer area of 9.425 m^2, a mass flow rate of 0.0590 kg/s and a sensible heat-transfer rate of 474.4 W.

The report covers the operating principle, literature context, design inputs, calculations, SOLIDWORKS modelling, ANSYS Workbench analysis, fabrication, experimental procedure, an illustrative time-series dataset and simulation-to-experiment comparison. The numerical data table is synthetic and included to show how measurements should be recorded and analysed; it is not represented as actual test data from the student's prototype.

Keywords: Earth Tube Heat Exchanger, EAHE, ground-coupled heat transfer, SOLIDWORKS, ANSYS Workbench, thermal analysis, fabrication, experimental validation.

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LIST OF SYMBOLS

Symbol	Description	Unit
A	Pipe cross-sectional area	m^2
A_s	Internal pipe heat-transfer area	m^2
c_p	Specific heat capacity of air	J/(kg·K)
D_i, D_o	Pipe internal and external diameters	m
f	Darcy friction factor	—
k	Thermal conductivity	W/(m·K)
L	Pipe length	m
$\dot{m}$	Air mass flow rate	kg/s
$\dot{Q}$	Sensible heat-transfer rate	W
T_in, T_out	Inlet and outlet air temperatures	°C
T_s	Soil temperature	°C
v	Mean air velocity	m/s
$\dot{V}$	Volumetric flow rate	m^3/s
ρ	Air density	kg/m^3
μ	Dynamic viscosity	Pa·s
Re	Reynolds number	—
Δp	Pressure drop	Pa
ε_T	Temperature-based effectiveness	—

CHAPTER 1: INTRODUCTION

1.1 Background

Energy consumption for ventilation and space conditioning is an important consideration in buildings. Conventional vapour-compression air conditioning provides controlled cooling but requires electrical energy to operate compressors, fans and other components. A ground-coupled heat exchanger offers an alternative or supplementary method of pre-conditioning ventilation air by using the ground as a thermal reservoir.

At an appropriate depth, soil temperature typically varies more slowly than outdoor air temperature over short time scales. A pipe buried in the soil provides a surface through which heat can transfer between the air inside the pipe and the surrounding ground. In summer, warmer air can reject heat to cooler soil; in winter, colder air can absorb heat from warmer soil. The magnitude of the temperature change depends on the soil profile, pipe dimensions, air velocity, moisture, operating time and thermal contact.

1.2 Problem statement

The design challenge is to obtain a useful temperature change while maintaining a practical airflow rate and acceptable pressure drop. A longer pipe may increase heat-transfer area, but it also increases friction losses and installation cost. Higher airflow increases the amount of air delivered but may reduce residence time and raise fan power. Therefore, the system should be assessed through both thermal and fluid-flow considerations.

1.3 Aim

To develop a design, analysis, fabrication and performance-evaluation methodology for an Earth Tube Heat Exchanger prototype.

1.4 Objectives

- Develop a dimensioned pipe layout using SOLIDWORKS.
- Identify the thermal and flow inputs required for preliminary design.
- Calculate cross-sectional area, mean velocity, mass flow, surface area and heat-transfer rate.
- Describe appropriate thermal and structural analysis workflows in ANSYS Workbench.
- Document fabrication, installation and sensor placement.
- Demonstrate how inlet/outlet temperature and airflow data are processed.
- Compare numerical predictions with measurements and discuss sources of uncertainty.

1.5 Scope

The scope covers preliminary design calculations, CAD modelling, thermal analysis methodology, relevant structural checks, fabrication considerations and experimental evaluation. The illustrative dataset is included to demonstrate data handling and is not a substitute for actual measurements. Final performance claims must be supported by verified test data.

1.6 Significance

The project connects thermodynamics, heat transfer, fluid mechanics, CAD/CAE and experimental engineering. It also demonstrates why design assumptions, material properties and boundary conditions must be documented so that numerical results can be compared fairly with experimental observations.

CHAPTER 2: LITERATURE REVIEW

2.1 Operating principle

An earth–air heat exchanger transfers heat through three principal mechanisms: convection between the moving air and inner pipe wall, conduction through the pipe wall, and conduction into or out of the surrounding soil. These

mechanisms act in series. The ground-side resistance can vary with soil properties, moisture, geometry and operating time.

2.2 Published research

Bisoniya, Kumar and Baredar reviewed experimental and analytical studies of EAHE systems and highlighted the role of soil and design conditions [1]. Bisoniya described a simplified design approach for earth–air heat exchangers [2]. Zhao and co-authors investigated design parameters using a model experiment [3]. Ahmad, Prakash and Saini combined three-dimensional simulation with experimental validation for heating-mode performance [4]. These studies support the use of site-specific data, engineering calculations and validation rather than relying on a single universal pipe dimension.

2.3 Parameters influencing performance

Parameter	Effect	Engineering response
Pipe length	Increases heat-transfer area; benefit may diminish as air approaches soil temperature	Optimise against pressure loss and cost
Internal diameter	Changes velocity, surface area and friction	Choose with target flow and fan curve
Burial depth	Affects soil temperature stability	Measure ground temperature at proposed depth
Soil conductivity and moisture	Affect ground-side heat transfer	Use site-specific or documented properties
Airflow	Changes mass flow and residence time	Measure and record during each test
Operating duration	Changes local soil temperature	Use time-series data; allow ground recovery
Bends and fittings	Increase local pressure losses	Include actual fittings in pressure-drop model
Condensation	May affect hygiene and performance	Plan slope, drainage and cleaning access

2.4 Research rationale

The project provides a structured workflow for relating design inputs to calculated performance and then to measured behaviour. A prototype-specific conclusion should be based on its geometry, soil conditions and experimental measurements rather than directly copying results from other climates or configurations.

CHAPTER 3: SYSTEM DESCRIPTION AND DESIGN METHODOLOGY

3.1 Illustrative design basis

For a complete worked example, this report assumes a straight pipe of 20 m length, internal diameter 0.150 m and external diameter 0.160 m. The air volumetric flow rate is assumed to be 0.050 m³/s. The warm-air inlet temperature is assumed to be 38°C, the outlet temperature used for calculations is 30°C, and the representative soil temperature is 25°C. These values are selected for demonstration and must not be described as actual prototype data.

Component/condition	Illustrative value	Purpose
Pipe length	20.0 m	Defines length and surface area
Pipe internal diameter	0.150 m	Defines flow area and internal surface
Pipe external diameter	0.160 m	Defines wall thickness
Airflow	0.050 m³/s	Sets volumetric ventilation flow
Inlet air temperature	38.0°C	Warm-weather inlet condition
Outlet air temperature	30.0°C	Assumed output for worked calculation
Soil temperature	25.0°C	Ground-side reference temperature
Air density	1.18 kg/m³	Mass flow calculation
Air specific heat	1005 J/(kg·K)	Sensible heat calculation
Pipe conductivity	0.40 W/(m·K)	Illustrative wall property
Soil conductivity	1.20 W/(m·K)	Illustrative soil property

3.2 System components

- Buried pipe: provides the heat-transfer surface between air and soil.
- Inlet section: admits ambient air and should limit entry of debris and water.
- Outlet section: conveys conditioned air to the test point or intended space.
- Fan (if used): supplies the pressure rise needed to overcome system resistance.
- Temperature sensors: measure inlet, outlet and soil temperatures.
- Airflow instrument: measures volumetric flow or local velocity.
- Drainage and access provisions: address condensation, inspection and cleaning.

3.3 Working sequence

1. Ambient air enters at T_in.
2. Air flows through the pipe at the selected flow rate.
3. Heat transfers between the air and pipe wall by convection.
4. Heat conducts through the pipe wall and the surrounding soil.
5. Air leaves at T_out; the actual outlet temperature is measured.
6. Temperature change, heat-transfer rate and pressure loss are calculated.

3.4 Design workflow

7. Define intended use, design airflow and climate conditions.
8. Measure or estimate soil temperature at the proposed burial depth.
9. Select preliminary pipe material, length, diameter and layout.
10. Calculate area, velocity, mass flow and heat-transfer surface.
11. Estimate pressure drop and assess fan requirements.
12. Create a CAD model and dimensioned drawing.
13. Set material properties, mesh and boundary conditions for simulation.
14. Fabricate and install the prototype with suitable access and drainage.
15. Conduct repeatable tests and compare results against calculations and simulation.

CHAPTER 4: DETAILED ENGINEERING CALCULATIONS

4.1 Input data and assumptions

Parameter	Symbol	Assumed value	Use
Internal diameter	D_i	0.150 m	Flow area
External diameter	D_o	0.160 m	Wall geometry
Length	L	20.0 m	Surface area and friction
Volumetric flow	$\dot{V}$	0.050 m³/s	Velocity and mass flow
Density	ρ	1.18 kg/m³	Mass flow
Specific heat	c_p	1005 J/(kg·K)	Sensible heat
Viscosity	μ	1.85×10^{-5} Pa·s	Reynolds number
Inlet temperature	T_{in}	38.0°C	Thermal condition
Outlet temperature	T_{out}	30.0°C	Assumed for calculation
Soil temperature	T_s	25.0°C	Ground reference
Friction factor	f	0.025	Illustrative pressure loss
Local-loss sum	ΣK	3.0	Illustrative fittings losses

4.2 Cross-sectional area

For a circular pipe, the internal flow area is $A = \pi D_i^2/4$.

$A = \pi \times (0.150\ \text{m})^2/4 = 0.017671\ \text{m}^2$.

4.3 Mean air velocity

The mean velocity is $v = \dot{V}/A$.

$v = 0.050\ m^3/s \div 0.017671\ m^2 = 2.829\ m/s$.

4.4 Internal heat-transfer area

For a straight cylindrical pipe, neglecting end surfaces, $A_s = \pi D_i L$.

$A_s = \pi \times 0.150\ m \times 20.0\ m = 9.425\ m^2$.

4.5 Mass flow rate

The air mass flow rate is $\dot{m} = \rho\dot{V}$.

$\dot{m} = 1.18\ kg/m^3 \times 0.050\ m^3/s = 0.0590\ kg/s$.

4.6 Temperature reduction

The temperature reduction is $\Delta T = T_{in} - T_{out}$.

$\Delta T = 38.0°C - 30.0°C = 8.0\ K$.

4.7 Sensible heat-transfer rate

For sensible heat transfer, $\dot{Q} = \dot{m}c_p(T_{in} - T_{out})$.

$\dot{Q} = 0.0590\ kg/s \times 1005\ J/(kg\cdot K) \times 8.0\ K = 474.4\ W = 0.474\ kW$.

This estimate assumes no latent heat contribution from condensation and uses constant approximate air properties. It is the heat removed from the air under the assumed data, not necessarily the useful cooling delivered to a room.

4.8 Temperature-based effectiveness

For a simplified case in which the ground is treated as a large reservoir at T_s, $\varepsilon_T = (T_{in} - T_{out})/(T_{in} - T_s)$.

$\varepsilon_T = (38 - 30)/(38 - 25) = 0.6154 = 61.54\%$.

This ratio indicates the fraction of the inlet-to-soil temperature difference achieved by the assumed outlet temperature. It is not the overall energy efficiency of the system.

4.9 Reynolds number

Reynolds number is $Re = \rho v D_i/\mu$.

$Re = 1.18 \times 2.829 \times 0.150/(1.85 \times 10^{-5}) = 27071$.

This value is above the usual approximate transition range for internal pipe flow, suggesting turbulent flow under the assumed conditions. Actual flow regime depends on the true flow, geometry and inlet conditions.

4.10 Pressure drop estimate

A simplified Darcy–Weisbach estimate with local losses is $\Delta p = [f(L/D_i) + \Sigma K](\rho v^2/2)$.

Dynamic pressure $= \rho v^2/2 = 1.18 \times 2.829^2/2 = 4.723\ Pa$.

Major loss $= 0.025 \times (20/0.150) \times 4.723 = 15.74\ Pa$.

Minor loss $= 3.0 \times 4.723 = 14.17\ Pa$.

Total illustrative pressure drop $= 15.74 + 14.17 = 29.91\ Pa$.

The estimate is sensitive to pipe roughness, actual bends, joints, entrance/exit losses, filter losses and the selected friction factor. It should not be used to select a fan without a complete system curve.

4.11 Air power associated with pressure drop

The ideal air power is $P_{air} = \dot{V}\Delta p$.

$P_{air} = 0.050\ m^3/s \times 29.91\ Pa = 1.496\ W$.

The electrical power drawn by a fan will be higher because of fan and motor inefficiencies. Fan input power should be measured for a complete energy assessment.

4.12 Pipe wall thermal resistance

For a cylindrical pipe wall, conduction resistance per unit length is $R'_{wall} = \ln(r_o/r_i)/(2\pi k_{pipe})$.

With $r_i = 0.075$ m, $r_o = 0.080$ m and $k_{pipe} = 0.40\ W/(m\cdot K)$, $R'_{wall} = 0.02568\ K\cdot m/W$.

This is only the pipe-wall contribution. The total resistance includes air-side convection, pipe-wall conduction, contact effects and soil conduction. Transient ground response may require a time-dependent model.

CHAPTER 5: CAD MODELLING AND ANSYS ANALYSIS

5.1 SOLIDWORKS modelling

For the illustrative model, create a hollow cylinder with 150 mm internal diameter, 160 mm outer diameter and 20,000 mm length. Add actual bends, connectors, supports and sensor ports from the prototype drawing. Generate an assembly view and dimensioned engineering drawing. The final report should contain genuine screenshots from the project model rather than generic images.

5.2 ANSYS thermal analysis workflow

16. Import or create the geometry and confirm unit consistency.
17. Assign pipe and soil properties from verified material data.
18. Select steady-state or transient thermal analysis according to the study objective.
19. Generate the mesh and refine it where temperature gradients are important.
20. Apply inlet temperature, soil boundary conditions, thermal contact and external heat-transfer conditions.
21. Solve and review temperature contours, wall heat flux and temperature variation along the pipe.
22. Perform mesh-sensitivity and energy-balance checks if feasible.
23. Compare predicted and measured outlet temperatures under matching operating conditions.

5.3 Thermal versus structural analysis

Thermal analysis predicts temperature distribution and heat flux. Structural analysis predicts stress, strain and deformation under specified mechanical or thermal loads. Structural analysis is meaningful only when actual loads and restraints are defined, for example soil overburden, external pressure, support forces or thermal expansion. Report material strength, load cases and boundary conditions with any stress result.

5.4 Illustrative simulation comparison

For demonstration only, assume ANSYS predicts an outlet temperature of 29.5°C while a test reads 30.0°C under comparable conditions. Percentage error is $|T_{sim} - T_{exp}|/|T_{exp}| \times 100$.

Percentage error = $|29.5 - 30.0|/30.0 \times 100 = 1.67\%$.

This example is not an actual validation result. A real comparison requires verified simulation and experimental values, consistent boundary conditions and consideration of instrument uncertainty.

CHAPTER 6: FABRICATION AND EXPERIMENTAL METHODOLOGY

6.1 Fabrication procedure

24. Verify pipe material, dimensions and fittings against the design drawing.
25. Measure and mark the pipe sections and joint positions.
26. Cut and prepare the ends using a method suitable for the selected material.
27. Assemble the pipe, fittings and inlet/outlet sections.
28. Inspect the joints, alignment and unobstructed internal passage.
29. Install the pipe at the chosen depth and protect it during backfilling.
30. Provide drainage and slope where condensation may occur.
31. Install sensors and airflow equipment at representative locations.
32. Check for leakage and confirm safe operation before testing.

6.2 Instrumentation

Instrument	Measurement	Details to record
Temperature sensors / thermocouples	Inlet/outlet air temperature	Type, accuracy, position and calibration
Soil temperature probe	Soil temperature near pipe	Depth and distance from pipe

Anemometer / flow meter	Airflow rate	Method, range and uncertainty
Differential pressure gauge (optional)	Pressure drop	Measurement points and range
Humidity sensor (optional)	Relative humidity	Sensor accuracy and location
Power meter (if fan is used)	Fan electrical input	Fan setting and input power

6.3 Experimental procedure

33. Record date, time, ambient conditions and soil temperature.
34. Measure inlet temperature and airflow.
35. Operate the fan at a documented setting if used.
36. Record outlet temperature at fixed intervals or after a defined stabilisation period.
37. Measure pressure drop and fan power if instruments are available.
38. Repeat trials under comparable conditions.
39. Calculate temperature reduction and heat-transfer rate for each observation.
40. Compare the data with simulation results using matching boundary conditions.

6.4 Data handling

The sample table below demonstrates time-series recording for a one-hour test at ten-minute intervals. All entries are synthetic and are included only to demonstrate analysis. They must not be claimed as measured data from the project.

Time (min)	Inlet (°C)	Outlet (°C)	Soil (°C)	Flow (m³/s)	ΔT (K)	Q̇ (W)
0	38.0	33.0	25.0	0.050	5.0	296.5
10	38.2	31.8	25.0	0.050	6.4	379.5
20	38.1	30.9	25.1	0.051	7.2	435.5
30	38.0	30.4	25.1	0.050	7.6	450.6
40	38.3	30.2	25.2	0.049	8.1	470.7
50	38.4	30.1	25.2	0.050	8.3	492.1
60	38.2	30.0	25.2	0.050	8.2	486.2

Mean illustrative inlet temperature = 38.17°C; mean outlet temperature = 30.91°C; mean soil temperature = 25.11°C; mean airflow = 0.0500 m³/s.

Mean illustrative temperature reduction = 7.26 K; mean sensible heat-transfer rate from the row-wise calculations = 430.2 W.

6.5 Safety and hygiene

Excavation should be preceded by checks for underground utilities and suitable trench safety. The inlet should be protected from debris, insects and surface water. Condensation should be managed with drainage and access for inspection. If air is supplied to occupied spaces, filtration, cleaning access and indoor-air-quality considerations are essential. Electrical components must be installed with appropriate protection.

CHAPTER 7: EXPERIMENTAL RESULTS AND DISCUSSION

7.1 Summary of calculated outputs

Metric	Illustrative result	Meaning
Cross-sectional area	0.017671 m²	Flow area
Mean velocity	2.829 m/s	Average velocity from volumetric flow
Internal surface area	9.425 m²	Straight-pipe heat-transfer area
Mass flow rate	0.0590 kg/s	Air mass passing through pipe per second
Temperature reduction	8.0 K	Assumed inlet-to-outlet difference
Sensible heat rate	474.4 W	Heat removed under assumed conditions

Temperature effectiveness	61.54%	Simplified temperature ratio
Reynolds number	27071	Flow regime indicator
Pressure drop	29.91 Pa	Estimate based on assumed loss coefficients
Air power	1.496 W	Ideal power associated with estimated pressure drop

7.2 Time-series interpretation

In the synthetic example, outlet temperature falls from 33.0°C at the start to 30.0°C at 60 minutes, while the inlet remains near 38°C and soil temperature remains close to 25°C. Such a trend could represent a system approaching a more stable thermal response, but it cannot establish the actual behaviour of a prototype. A real test should consider sensor response, changing ambient conditions and local soil warming.

The row-wise heat-transfer estimate is calculated from each recorded inlet temperature, outlet temperature and flow rate. This is preferable to assuming a constant flow when the flow changes between observations. For a formal report, uncertainty in airflow and temperature measurement should also be considered.

7.3 Comparison with illustrative simulation

Parameter	Illustrative simulation	Illustrative experimental value	Difference
Outlet temperature	29.5°C	30.0°C	0.5°C
Percentage error	—	—	1.67%
Temperature drop for 38°C inlet	8.5 K	8.0 K	0.5 K

The example percentage error is 1.67%. This is an arithmetic demonstration only and should not be cited as the actual model validation error. Real validation should match airflow, inlet temperature, soil condition, pipe geometry and operating time as closely as possible.

7.4 Sources of uncertainty

- Temperature sensor accuracy, calibration and placement.
- Airflow measurement error and non-uniform velocity profile.
- Variation in soil moisture and conductivity.
- Heat gain in exposed inlet/outlet sections.
- Air leakage at joints.
- Uncertainty in pipe material properties and thermal contact.
- Simplifications in steady-state or boundary-temperature assumptions.
- Pressure losses from bends, filters and fittings omitted from the model.

7.5 Engineering implications

The calculations show that outlet temperature reduction alone is insufficient to judge a ground-coupled system. Airflow, heat-transfer rate, pressure drop and fan power should be considered together. A design that achieves a large temperature reduction at very low airflow may deliver limited ventilation, while high airflow may increase fan energy and reduce the temperature change per unit mass. The final design should be selected according to the required application and measured operating conditions.

CHAPTER 8: CONCLUSIONS AND FUTURE SCOPE

8.1 Conclusions

An Earth Tube Heat Exchanger uses the ground as a thermal reservoir to pre-condition ventilation air. The principal variables include the air-to-soil temperature difference, pipe dimensions, airflow, soil thermal properties, pressure

drop and operating duration. A sound engineering assessment combines CAD geometry, heat-transfer calculations, simulation with documented boundary conditions, safe fabrication and repeatable experiments.

For the illustrative input set, the calculated internal area is 0.017671 m², mean velocity is 2.829 m/s, mass flow rate is 0.0590 kg/s and sensible heat-transfer rate is 474.4 W. The estimated pressure drop is 29.91 Pa using assumed friction and local-loss factors. These results demonstrate the calculation method and are not verified performance values for the actual prototype.

The project-specific conclusion must be finalised using the actual design dimensions, simulation output and experimental measurements. No measured energy saving or prototype cooling capacity should be claimed without supporting data.

8.2 Future scope

- Measure the soil temperature profile and thermal properties at the installation site.
- Investigate several pipe lengths, diameters, burial depths and flow rates.
- Use transient soil analysis or coupled CFD where suitable and supported by data.
- Improve test instrumentation and repeat measurements across multiple days.
- Measure fan power and evaluate total system energy performance.
- Investigate condensation management, filtration, cleaning access and air quality.
- Compare pipe materials and configurations using a cost-performance assessment.

APPENDIX A: FULL ILLUSTRATIVE TIME-SERIES DATA

The following data are synthetic and must not be presented as actual observations. They are provided so the report contains a complete example of recording and processing experimental data.

Time (min)	T_in (°C)	T_out (°C)	T_soil (°C)	$\dot{V}$ (m³/s)	ΔT (K)	$\dot{m}$ (kg/s)	$\dot{Q}$ (W)
0	38.0	33.0	25.0	0.050	5.0	0.0590	296.5
10	38.2	31.8	25.0	0.050	6.4	0.0590	379.5
20	38.1	30.9	25.1	0.051	7.2	0.0602	435.5
30	38.0	30.4	25.1	0.050	7.6	0.0590	450.6
40	38.3	30.2	25.2	0.049	8.1	0.0578	470.7
50	38.4	30.1	25.2	0.050	8.3	0.0590	492.1
60	38.2	30.0	25.2	0.050	8.2	0.0590	486.2

APPENDIX B: ASSUMPTIONS AND SUBMISSION CHECKLIST

Assumptions used in this worked example: straight 20 m pipe; 0.150 m internal diameter; 0.160 m outside diameter; constant air density and specific heat; inlet temperature around 38°C; soil temperature around 25°C; constant or near-constant airflow; illustrative outlet temperatures; assumed friction factor and local-loss coefficient; no latent heat contribution; no claim of measured energy saving.

- Replace the pipe dimensions and material with the actual prototype specification.
- Replace synthetic time-series data with the original laboratory readings.
- Insert real SOLIDWORKS screenshots, dimensioned drawings and prototype photographs.
- Insert real ANSYS mesh, boundary conditions, temperature contours and structural outputs.
- Document instrument type, accuracy, calibration and measurement location.
- Verify the literature references from publisher or institutional-library records.
- Add the supervisor name and follow the department's prescribed report format.
- Ensure all statements about fabrication, testing and validation reflect the work actually completed.

REFERENCES

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[4] S. N. Ahmad, O. Prakash and P. Saini, “Performance investigation of earth air heat exchanger: A synergistic approach of 3D simulation and its experimental validation for heating mode,” Environmental Progress & Sustainable Energy, vol. 44, no. 4, e14620, 2025. DOI: 10.1002/ep.14620.

Consult the full articles and verify all bibliographic details before final submission. Add the exact software versions, material datasheets and other sources actually used.